1. rev — Reverse each line

```
vanam@Uday:~$ echo "linux" | rev
xunil
```

2. tac — Read a file backwards

```
vanam@Uday:~$ vim file1
```

```
first
Second
Third

~
~
~
~
~
~
~
~
~
~
~
```

```
vanam@Uday:~$ tac file1

Third
Second
first
```

3. fold — Wrap long lines

- -w 10 — wrap at 10 characters.
- -s — prefer breaking at spaces.

```
vanam@Uday:~$ echo "Linux commands are useful" | fold -w 10
echo "Linux commands are useful" | fold -s -w 10
Linux comm
ands are u
seful
Linux
commands
are useful
```

**4. expand — Convert tabs to spaces**

-t 4 — set tab stops every four columns.

```
vanam@Uday:~$ printf 'Java\tSpring\n' > file1
expand -t 4 file1
Java    Spring
```

**5. unexpand — Convert spaces to tabs**

- -a — convert all suitable groups of spaces, not just leading spaces.

```
vanam@Uday:~$ printf '        Hello Linux\n' > file1
vanam@Uday:~$ unexpand -a file1
        Hello Linux
```

**6. nl — Number lines**

- -ba — number all lines, including blank lines.
- -w 3 — use a three-character-wide number field.

```
vanam@Uday:~$ printf 'Java\n\nPython\n' > file1
nl -ba file1
nl -ba -w 3 file1
     1  Java
     2
     3  Python
  1     Java
  2
  3     Python
```

**7. paste — Combine lines side by side**

```
vanam@Uday:~$ printf 'Uday\nRavi\n' > names.txt
printf 'Java\nPython\n' > skills.txt
paste names.txt skills.txt
Uday    Java
Ravi    Python
```

**8. join — Combine matching records**

```
vanam@Uday:~$ printf '1 Uday\n2 Ravi\n' > names.txt
printf '1 Java\n2 Python\n' > skills.txt
join names.txt skills.txt
1 Uday Java
2 Ravi Python
```

**9. comm — Compare two sorted files**

```
vanam@Uday:~$ printf 'Java\nPython\nSQL\n' > file1
printf 'Java\nLinux\nPython\n' > file2
comm file1 file2
                Java
        Linux
                Python
SQL
```

**10. split — Break a file into smaller files**

**Options:**

- **-l 2 — put two lines in each output file.**
- **-b 1K — split into chunks of approximately 1 KB.**
- **-d — use numeric suffixes.**

```
vanam@Uday:~$ seq 1 10 > numbers.txt
split -l 3 -d numbers.txt chunk_
ls chunk_*
cat chunk_00
chunk_00  chunk_01  chunk_02  chunk_03
1
2
3
```